

Energy: stores, transfers and work

From a braking car to a law of nature

Physics · 2A · 21 September 2026

A car brakes hard on a dry road

(slide 2)

At about 50 km/h (30 mph) a car stops in about 14 m. These are brakes only: the driver's reaction time is not included.

Before moving on. And at about 100 km/h (60 mph)? About 28 m, about 40 m or about 55 m? Write your choice down, with the reason: we do not solve it now.

The question stays open until the end of the lesson: the work we are about to do is the price of the answer.

Source: The Highway Code (UK), typical braking distances: 14 m at 30 mph, 55 m at 60 mph. Speeds in km/h are rounded. Typical values for a dry road and good brakes.

Stores and transfers of energy

(slide 3)

To follow the energy of the car we need a vocabulary, in the wording of Cambridge IGCSE: **store** and **transfer**. Energy is not “of a type”: it is stored somewhere, and it moves.

Energy stores (nouns)	Energy transfers (verbs)
kinetic	mechanically, by a force
gravitational potential	electrically, by a current
elastic (strain)	by heating
chemical	by waves (light, sound)
thermal (internal)	
nuclear	

A store holds energy. A transfer moves it from one store to another. An everyday example of each store: kinetic, a moving car; gravitational potential, a book on a shelf; elastic, a stretched rubber band; chemical, fuel, food, a battery; thermal, a hot cup; nuclear, the Sun.

Before moving on. Which word describes the car's engine: a store or a transfer?

Neither: the engine moves energy from the chemical store of the fuel to the kinetic store of the car.

A flow diagram: the braking car

(slide 4)

Where does the energy of the car come from, and where does it end up? We draw it as a flow diagram (figure 1).

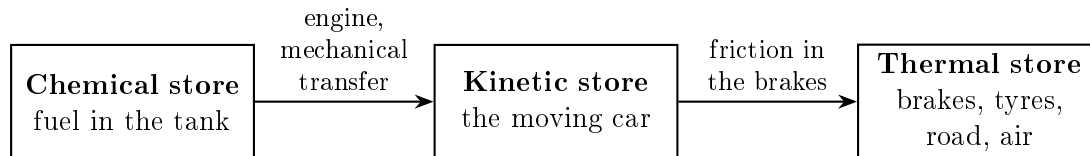


Figure 1: Flow diagram of the braking car: the energy of the car goes from the chemical store of the fuel, through the kinetic store, to the thermal store.

The car's kinetic energy is not “used up”: it becomes thermal energy in the brakes. The brakes get hot. The brake discs are hot after a hard stop: the energy is in the thermal store of the discs, pads, tyres, road and air. The chemical store is only at the start of the chain; we do not follow it further today.

The car is our thread for the whole lesson: the brakes must take away all of its kinetic energy.

Your turn: draw the flow diagram

(slide 5)

Before moving on. Draw on paper the flow diagram of a ball that bounces and comes back lower each time, and of a hydroelectric dam. Where does the missing height of the ball go?

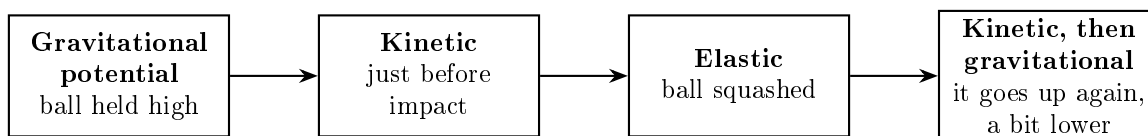


Figure 2: Flow diagram of a bouncing ball that comes back lower each time.

At each impact a small share of the energy goes to the thermal store (ball, floor, air) and to sound. The missing height is not destroyed.

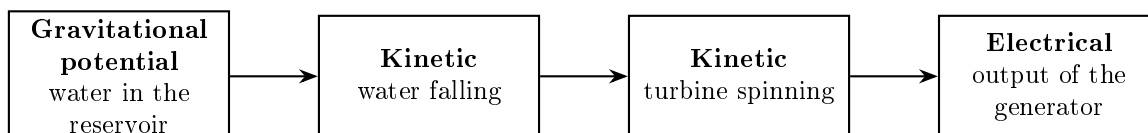


Figure 3: Flow diagram of a hydroelectric dam.

In the dam friction and the heating of the wires send a share to the thermal store: that share is small, but never zero. The same words, store and transfer, describe a machine, not just a car.

The law of conservation of energy

(slide 6)

Law. Energy cannot be created or destroyed. It can only be transferred from one store to another.

$$\text{total energy before} = \text{total energy after}$$

“Wasted” energy is not gone: it is in the thermal store of the surroundings, spread out and hard to use again. “Lost” is the wrong word; “wasted” (not useful) is the right one. We measure energy in joules (J).

From now on every calculation of the lesson is a bookkeeping of stores: what goes down, what goes up.

Before the first formula

(slide 7)

Before moving on. A car goes at 50 km/h. What is that in m/s? What is the weight of a 2 kg mass, with $g = 10 \text{ N/kg}$?

$50 \div 3.6 = 13.9 \text{ m/s}$, since $1 \text{ m/s} = 3.6 \text{ km/h}$. And $W = mg = 2 \times 10 = 20 \text{ N}$: mass is in kg, weight is a force, in N. In this course $g = 10 \text{ N/kg}$, the value used in IGCSE papers.

Kinetic energy

(slide 8)

Before moving on. What should the energy of a moving object depend on?

On its mass and on its speed. The formula says how:

Definition. The energy stored in a moving object.

$$E_k = \frac{1}{2} m v^2$$

E_k is the kinetic energy, in joules (J); m is the mass of the object, in kilograms (kg); v is the speed of the object, in metres per second (m/s), squared.

Every symbol must be defined before using it: v is the speed in m/s, not in km/h. Note the two features of the formula: the $\frac{1}{2}$ and the square on the speed. The next slide shows what the square does.

What does the square do?

(slide 9)

Before moving on. In figure 4, at 27.8 m/s (100 km/h), where will the curve be compared with the line?

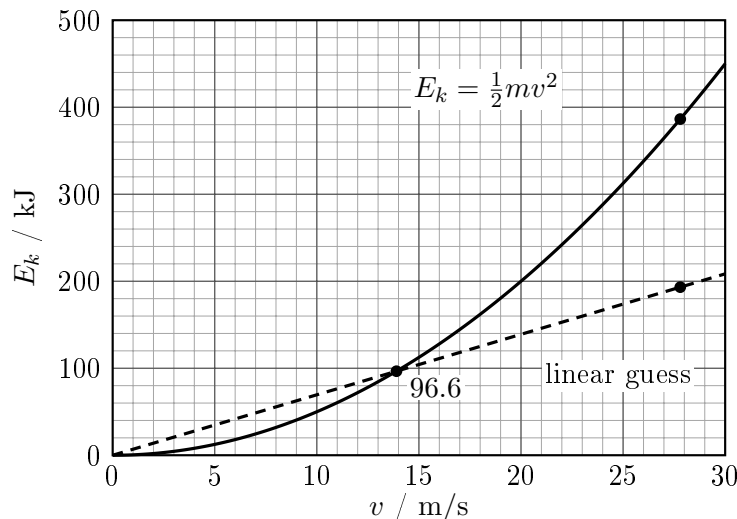


Figure 4: Kinetic energy of a car of 1 000 kg against speed: $E_k = \frac{1}{2}mv^2$ (solid) and the linear guess E_k proportional to v (dashed). The two curves agree at 13.9 m/s (50 km/h).

For a car of 1 000 kg the line gives 193 kJ and the curve gives 386 kJ: “double the speed, double the energy” is the straight line, our intuition; “double the speed, four times the energy” is the parabola. If the mass changes, the curve stretches vertically in proportion to m : double the mass, double the energy. The shape does not change: it is always a parabola.

A worked example: the braking car

(slide 10)

A car of mass 1 000 kg goes at 50 km/h, then at 100 km/h. What is its kinetic energy in each case?

1. $v_1 = \frac{50}{3.6} = 13.9$ m/s, $v_2 = \frac{100}{3.6} = 27.8$ m/s: first the units, speed must be in m/s.
2. $E_{k1} = \frac{1}{2} \times 1000 \times 13.9^2 \approx 96\,600$ J ≈ 97 kJ: then the law.
3. $E_{k2} = \frac{1}{2} \times 1000 \times 27.8^2 \approx 386\,000$ J ≈ 386 kJ: same calculation, other speed.
4. $\frac{E_{k2}}{E_{k1}} \approx 4$: double the speed, the brakes must remove four times the energy.

This is the first half of the answer to the question we left open. The brakes must remove four times the energy. Does that mean four times the distance? Hold that thought: we first need to know how a force takes energy away. That comes with the work, at the end.

Three typical mistakes

(slide 11)

Take the car at 50 km/h and get one thing wrong at a time.

- **Speed in km/h.** $\frac{1}{2} \times 1000 \times 50^2 = 1\,250\,000$ J: about 13 times too big, and it looks reasonable.
- **Forgetting the square.** $\frac{1}{2} \times 1000 \times 13.9 = 6\,950$ J: it only grows in proportion to v , not to v^2 .
- **Forgetting the half.** $1000 \times 13.9^2 = 193\,000$ J: exactly double the right answer.

Check every answer: is the unit the joule, and is the speed in m/s? This is the exact kind of mark lost in Paper 4.

Quick check

(slide 12)

Before moving on. The speed of a cyclist goes from 4 m/s to 12 m/s, same mass. The kinetic energy becomes 3 times bigger, 6 times bigger or 9 times bigger?

9 times bigger. The speed is multiplied by 3, so the energy is multiplied by $3^2 = 9$. Check: at 4 m/s, $E_k = \frac{1}{2}m \times 16 = 8m$; at 12 m/s, $E_k = \frac{1}{2}m \times 144 = 72m$; and $72/8 = 9$.

Your turn 1: kinetic energy

(slide 13)

Write: data, unit conversion, formula, substitution, result with unit.

- A ball of mass 0.50 kg moves at 8.0 m/s. Calculate its kinetic energy.
- The ball is now thrown at 16 m/s. What is its kinetic energy? How does it compare with (a)?
- A cyclist and bike, total mass 75 kg, travel at 36 km/h. Calculate the kinetic energy.

The solutions are at the end of the chapter.

Gravitational potential energy

(slide 14)

Definition. Lifting an object increases the energy stored in its gravitational field.

$$\Delta E_p = m g \Delta h$$

ΔE_p is the change in gravitational potential energy (J); m is the mass (kg); g is the gravitational field strength, 10 N/kg on Earth; Δh is the change in height (m).

Here h is the height and Δh is the *change* in height, measured from wherever the object starts. Recall that mg is the weight, in newtons: multiplied by a distance in metres it gives joules. We will see why in twenty minutes.

Only the change in height counts

(slide 15)

A 2.0 kg book goes from the table (0.75 m above the floor) to a shelf (1.95 m above the floor). Calculate ΔE_p (figure 5).

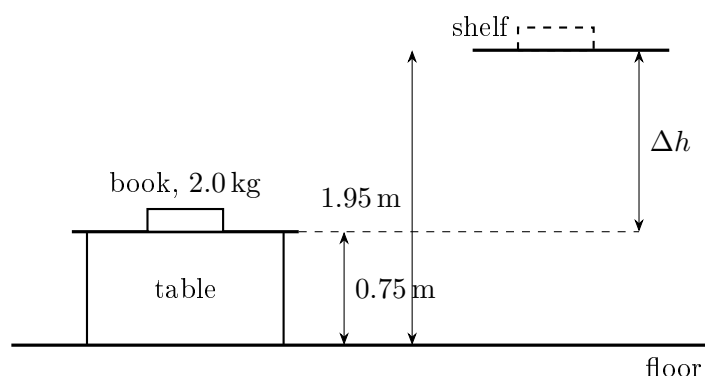


Figure 5: A 2.0 kg book goes from the table to a shelf. The heights above the floor are 0.75 m and 1.95 m; the change in height is Δh (schematic).

1. $\Delta h = 1.95 - 0.75 = 1.20$ m.
2. $\Delta E_p = 2.0 \times 10 \times 1.20 = 24$ J.

Not $2.0 \times 10 \times 1.95 = 39$ J: the floor is not where the book started. The energy *difference* does not depend on where we call “zero”. The path does not matter either: straight up or by a ramp, the same Δh gives the same ΔE_p .

Conservation at work: a falling ball

(slide 16)

A ball of mass 0.20 kg is dropped from 1.8 m. How fast is it going just before it hits the floor? Ignore air resistance.

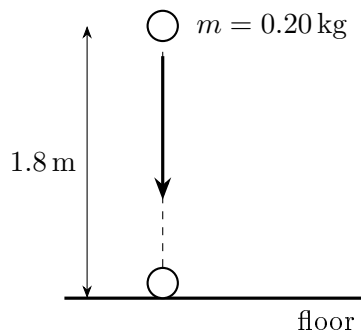


Figure 6: A ball dropped from 1.8 m (schematic).

Before moving on. Is a heavier ball faster?

1. $\Delta E_p = 0.20 \times 10 \times 1.8 = 3.6 \text{ J}$: the gravitational store goes down by 3.6 J.
2. $E_k = 3.6 \text{ J}$: conservation, the kinetic store goes up by the same amount.
3. $\frac{1}{2} \times 0.20 \times v^2 = 3.6 \Rightarrow v^2 = 36 \Rightarrow v = 6.0 \text{ m/s}$: solve for the speed, about 22 km/h.

No: with a ball twice as heavy the speed is the same, because the mass cancels. In symbols, $mg\Delta h = \frac{1}{2}mv^2$ and m drops out. We used only conservation and the two formulae; acceleration is not needed.

Your turn 2: potential energy and conservation

(slide 17)

- (a) A climber of mass 60 kg goes up a mountain path and gains 200 m in height. Calculate the increase in gravitational potential energy.
- (b) A 0.10 kg stone is thrown straight up at 12 m/s. How high above the hand does it go? Ignore air resistance.

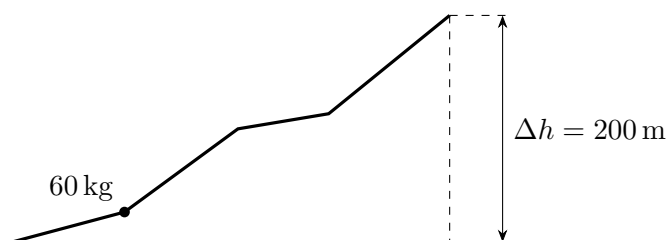


Figure 7: Exercise 2(a): the climber gains 200 m in height (schematic).

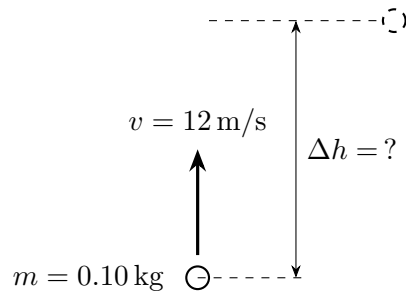


Figure 8: Exercise 2(b): the stone is thrown straight up at 12 m/s (schematic).

Part (b) is the reverse of the falling ball: kinetic energy goes into gravitational potential energy. The solutions are at the end of the chapter.

How does a force move energy?

(slide 18)

Before moving on. The brakes take energy out of the car. What are they doing, in physics words?

A force, friction, acts on the disc while it turns: the brakes do work.

Definition. When a force moves an object, it transfers energy to it. The energy transferred is the work done.

$$W = F d$$

W is the work done, equal to the energy transferred, in joules ($1 \text{ J} = 1 \text{ N} \times 1 \text{ m}$); F is the force, in newtons (N); d is the distance moved *in the direction of the force*, in metres (m).

Work is not a store: it is the name of the energy transfer by a force.

The potential energy formula was a work formula

(slide 19)

Lift the 2.0 kg book from the table to the shelf, slowly and at constant speed: $\Delta h = 1.20$ m (figure 9).

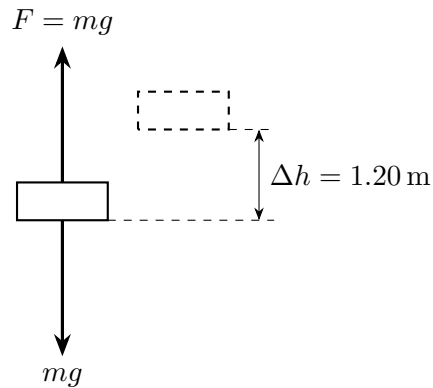


Figure 9: The book is lifted slowly, at constant speed: the lifting force F equals the weight mg (schematic).

1. $F = mg = 2.0 \times 10 = 20$ N: at constant speed the forces balance, so the lifting force equals the weight.
2. $W = Fd = 20 \times 1.20 = 24$ J: the hand transfers 24 J to the book.
3. $\Delta E_p = mg \Delta h = 24$ J: the same number. The gravitational store gains exactly what the hand transfers.

$mg \Delta h$ is just Fd with $F = mg$ and $d = \Delta h$. The chain is: hand (chemical store of the body), work by the lifting force, gravitational store of the book.

Back to the car

(slide 20)

The brakes exert an average force of about 7 000 N (dry road, good tyres). The work done by the brakes equals the kinetic energy removed (figure 10).

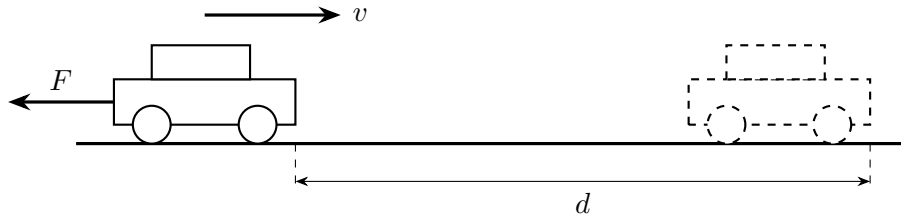


Figure 10: The brakes exert a force F opposite to the velocity v over a distance d (schematic).

1. $Fd = E_k \Rightarrow d = \frac{E_k}{F}$: the brakes stop the car when all the kinetic energy has been transferred.
2. 50 km/h: $d = \frac{96\,600}{7000} \approx 14$ m.
3. 100 km/h: $d = \frac{386\,000}{7000} \approx 55$ m.

Same force, four times the energy to remove: **four times the distance**. Both values agree with the Highway Code: 14 m and 55 m.

The value 7 000 N (about 0.7 of the weight of the car, 10 000 N) is an average chosen to match a dry road: it is data, not something we can compute yet. And the force is assumed constant; a real stop has small variations. The reaction (thinking) distance is not included: in the Highway Code it doubles, from 9 m to 18 m, while the braking distance quadruples.

Back to the class prediction

(slide 21)

At about 100 km/h (60 mph) the braking distance is not double: it is about **four times** that at 50 km/h. Kinetic energy grows with v^2 .

The intuition “double the speed, double the distance” is not silly: it is what a straight-line model says. The physics of energy says otherwise.

Braking distance, speed and braking force

(slide 22)

Before moving on. If the road gets wet, what changes in the formula $d = E_k/F$? What happens to d at 100 km/h?

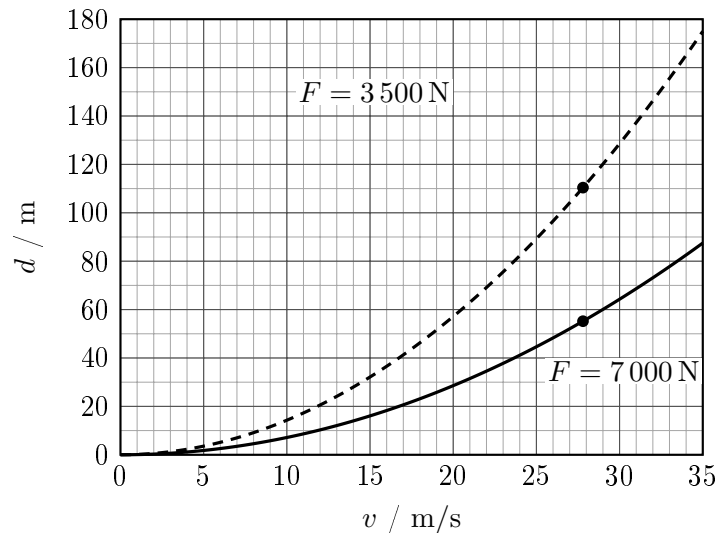


Figure 11: Braking distance $d = E_k/F$ against speed for a car of 1000 kg, for two values of the braking force. The dots mark 27.8 m/s (100 km/h).

F gets smaller, so d gets bigger. For a car of 1000 kg at 27.8 m/s: with $F = 7000$ N, $d \approx 55$ m; with half the force, $F = 3500$ N, $d \approx 110$ m, twice as far. The curve is a parabola because d is proportional to v^2 : the same reason as for E_k .

Your turn 3: work

(slide 23)

- A crate of mass 40 kg is pushed 5.0 m along the floor by a horizontal force of 120 N. How much work is done by the force?
- You carry a 20 kg bag horizontally for 10 m at constant speed. How much work does your upward force do on the bag?
- A cyclist and bike, total mass 80 kg, travel at 6.0 m/s and stop with a braking force of 120 N. What distance do they need? And if they were going at 12 m/s?

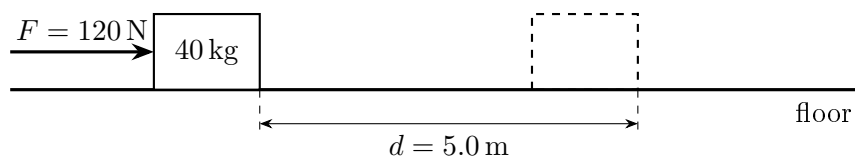


Figure 12: Exercise 3(a): the crate is pushed along the floor by a horizontal force (schematic).

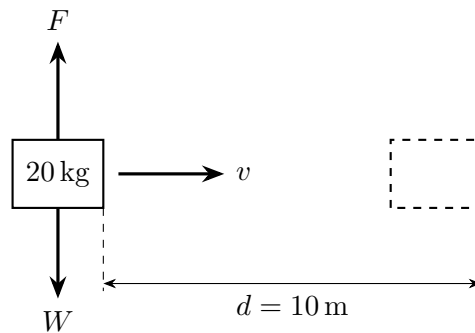


Figure 13: Exercise 3(b): the bag is carried horizontally at constant speed; F is the upward force on the bag, W its weight (schematic).

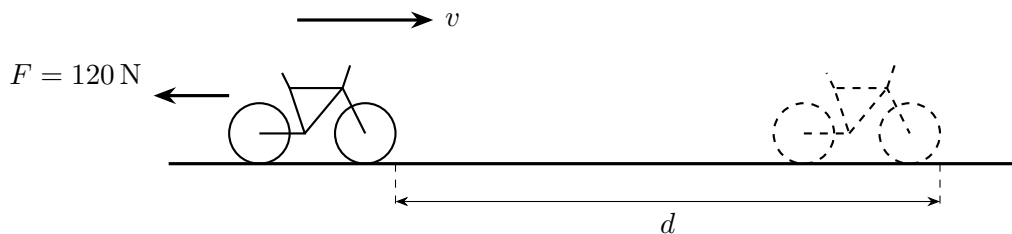


Figure 14: Exercise 3(c): the braking force F acts against the velocity v over the stopping distance d (schematic).

Part (c) is the problem of the hook, on a bicycle. The solutions are at the end of the chapter.

For 2 October: your climb

(slide 24)

The product to hand in is a table with your own numbers, written by hand.

1. Measure your mass (bathroom scale) and calculate your own weight in N ($g = 10 \text{ N/kg}$).
2. Measure the height of three different steps of a staircase with a ruler, to the nearest half division. Write all three values. Take the mean step height.
3. Count the steps of one flight: $\Delta h = \text{number of steps} \times \text{mean height}$.
4. Calculate ΔE_p for climbing that flight, and the work $W = Fd$ done by your leg force. Are they the same?

Everyone gets a different answer. In the ideal case, at constant speed, F is your weight and $d = \Delta h$, so $\Delta E_p = mg \Delta h$ and $W = Fd$ are equal. The three step heights will not agree: reading error, uneven steps.

In summary

(slide 25)

1. Energy is stored in stores and moved by transfers: it is conserved, never lost.
2. Kinetic energy: $E_k = \frac{1}{2}mv^2$. Double the speed, four times the energy.
3. Gravitational potential energy: $\Delta E_p = mg \Delta h$. Only the change in height counts.
4. Work: $W = Fd$ is the energy a force transfers. d is measured in the direction of the force.
5. Braking: $d = E_k/F$. That is why the distance is four times longer.

Solutions to the exercises

(slide 13, 17, 23)

Your turn 1

- (a) $E_k = \frac{1}{2} \times 0.50 \times 8.0^2 = \frac{1}{2} \times 0.50 \times 64 = 16 \text{ J}$.
- (b) $E_k = \frac{1}{2} \times 0.50 \times 16^2 = 64 \text{ J}$: four times the value in (a), because the speed has doubled.
- (c) $36 \text{ km/h} = 36 \div 3.6 = 10 \text{ m/s}$. $E_k = \frac{1}{2} \times 75 \times 10^2 = 3750 \text{ J} = 3.75 \text{ kJ}$.

Your turn 2

- (a) $\Delta E_p = mg \Delta h = 60 \times 10 \times 200 = 120\,000 \text{ J} = 120 \text{ kJ}$.
- (b) Kinetic energy at the start: $E_k = \frac{1}{2} \times 0.10 \times 12^2 = 7.2 \text{ J}$. At the top all of it is in the gravitational store: $7.2 = 0.10 \times 10 \times \Delta h$, so $\Delta h = 7.2 \text{ m}$.

Your turn 3

- (a) $W = Fd = 120 \times 5.0 = 600 \text{ J}$. The mass is not needed.
- (b) $W = 0 \text{ J}$: the upward force is at right angles to the displacement, so the distance moved in the direction of the force is zero. (Your muscles still use energy, but that is not work done on the bag.)
- (c) At 6.0 m/s : $E_k = \frac{1}{2} \times 80 \times 6.0^2 = 1\,440 \text{ J}$; $d = E_k/F = 1440/120 = 12 \text{ m}$. At 12 m/s : $E_k = \frac{1}{2} \times 80 \times 12^2 = 5\,760 \text{ J}$; $d = 5760/120 = 48 \text{ m}$: four times again.